

Paper XI — The Baryon Sector: The Colour-Singlet Proton, Centre-Phase Closure, the Dynamical Pion, and the First Nucleus

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Series: “A Universe in a Bottle” — Paper XI of XI. Specialises the fermion sector (Paper IX) and the strong force (Paper V) to colour-singlet bound states; uses the NJL constituent-mass machinery of Paper III.

New for the eleven-paper series (2026-06-08): the baryon sector is given a dedicated, full treatment — the colour-singlet proton, the centre-phase closure principle, the dynamical pion, the deuteron, and the derived NN repulsive core.

Abstract

We assemble the baryon sector of the BCC quantum-cellular-automaton model: how confined colour (Paper V) produces colourless, bound, stable matter, from the single proton up to the first nucleus. Five results organise it. First, the proton is the colour-singlet interpolating operator $B = \varepsilon_{abc} u^a u^b d^c$, gauge-invariant because $\det V = 1$ for $SU(3)$, with exactly zero colour charge ($C_2 = 0$), the correct quantum numbers ($Q = +1$, $B = 1$, $T_3 = \frac{1}{2}$), a fully antisymmetric total wavefunction (Fermi statistics via colour- $\varepsilon \times$ spin-flavour-symmetric), and energetic binding from the exact string tension. Second, the baryon’s stability is governed by a **centre-phase closure principle**: a colour state is asymptotic iff its $\mathbb{Z}_3$ centre-phase budget closes (N-ality 0), and a no-go theorem proves the two-body bound-pair fixed point does *not* extend to three constituents — which *predicts* that baryon mass is $\sim 99\%$ field energy (the PDG quark sum is 0.96% of m_p), the object enforcing the budget (the colour-dielectric condensate) being itself the binder. Third, the pion is delivered as a dynamical $q\bar{q}$ pseudoscalar Goldstone: massless in the chiral limit (exact), GMOR-linear at the physical point, and reproducing the measured m_π , f_π , $\langle\bar{q}q\rangle$ to $0.2\text{--}4\%$ from a single NJL coupling. Fourth, the deuteron — the first nucleus — binds as a single shallow $J^P = 1^+$, $I = 0$ state **only** through the pion tensor force, with the exact Rarita-Schwinger tensor strength and the physical $E_b = 2.224$ MeV, $\kappa = 0.2316$ fm $^{-1}$. Fifth, the short-range NN repulsive core is *derived* from the model’s own algebra — quark Pauli antisymmetry plus the chromomagnetic interaction give $+341.8$ MeV at zero separation, with the one external number (g_{cm}) fixed by the same $N\text{--}\Delta$ splitting the model already uses. All structural results are verified to machine precision or exact rationals.

1. Introduction

Baryons are the colour-singlet three-quark bound states; the proton and neutron are the lightest and the building blocks of nuclei. The strong force (Paper V) confines colour and makes the singlet the only finite-energy configuration, but confinement alone does not deliver a proton, a pion, or a deuteron — those are *composite* objects

with definite quantum numbers, statistics, masses, and a residual nuclear force. This paper builds them, following the project's matter-binding roadmap from the single hadron (the proton) up to the first nucleus (the deuteron).

The treatment is layered. Section 2 builds the proton as a colour-singlet operator with the right symmetry and an energetic binding argument. Section 3 states the closure principle that governs which colour combinations are stable and *why baryon mass is field energy, not constituent rest mass*. Sections 4–6 build the residual nuclear force from the bottom up: the dynamical pion (the long-range carrier), the deuteron it binds (through the tensor force), and the short-range repulsive core derived from quark substructure. Throughout, the constituent-mass scale and the chiral condensate are inherited from the NJL machinery of Paper III.

2. The colour-singlet proton

2.1 The interpolating operator

The first composite hadron is the colour-singlet baryon operator

$$B(x) = \varepsilon_{abc} q_1^a(x) q_2^b(x) q_3^c(x), \quad (q_1, q_2, q_3) = (u, u, d) \text{ for the proton,}$$

with the totally antisymmetric $SU(3)$ colour tensor ε_{abc} . Under a local gauge rotation $q^a \rightarrow V^a_{a'} q^{a'}$ with $V \in SU(3)$,

$$B \rightarrow \varepsilon_{abc} V^a_{a'} V^b_{b'} V^c_{c'} q^{a'} q^{b'} q^{c'} = \det(V) B = B,$$

because $\det V = 1$. The proton is therefore exactly colour-neutral (Finding F71, residual 1.2×10^{-15}). The ε contraction is the unique singlet of $3 \otimes 3 \otimes 3 = 1 \oplus 8 \oplus 8 \oplus 10$; the normalised singlet $|S\rangle = \frac{1}{\sqrt{6}} \varepsilon_{abc} |abc\rangle$ is annihilated by every total colour generator $G^a = \sum_i T^a_{(i)}$, with quadratic Casimir $C_2|S\rangle = 0$ (vs $4/3$ for a single quark, 3 for the octet) to 1.4×10^{-16} .

2.2 Quantum numbers and Fermi statistics

The proton uud has $Q = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1$, baryon number $B = \frac{1}{3} \cdot 3 = 1$, $T_3 = \frac{1}{2}$, all exact via rational arithmetic, with Gell-Mann-Nishijima $Q = T_3 + Y/2$ per quark ($Y = \frac{1}{3}$) consistent with the anomaly-free assignment of Paper IX. Fermi statistics are made explicit: with two identical u colour-triplets, $\varepsilon_{abc} u^a u^b d^c \equiv 0$ unless the rest of the wavefunction supplies the matching symmetry, so the total wavefunction factorises as

$$\Psi = \underbrace{(\text{colour } \varepsilon)}_{\text{antisymmetric}} \otimes \underbrace{(\text{spin-flavour})}_{\text{symmetric}} \otimes \underbrace{(\text{space, ground})}_{\text{symmetric}}.$$

The proton spin-up $SU(6)$ spin-flavour wavefunction (the 56-plet symmetric combination, 9 terms) is built explicitly and verified fully S_3 -symmetric (bit-for-bit). Under any quark transposition the colour part flips sign while spin-flavour and space are even, so $\Psi \rightarrow -\Psi$: the proton obeys Fermi statistics exactly.

2.3 Energetic binding

Using the exact 2D string tension (Paper V), the cost to pull one quark a distance R out of the singlet is $V(R) = \sigma(\beta)R$, strictly linear and unbounded (at $\beta = 2.0$, $\sigma = 2.051$, giving $V = 2.05, 4.10, 8.20, \dots$). The colour singlet is the finite-energy configuration; an isolated quark costs infinite energy. Confinement ($\sigma > 0$) plus the colour singlet together answer both halves of “why no free quarks, and why a proton exists.” This is an operator-level construction with an energetic binding argument; a real-time three-body flux-tube bound state (and the calibrated proton mass) is roadmap phase P2, not claimed here.

3. The centre-phase closure principle and the origin of baryon mass

3.1 Closure governs stability

The pair sector (Paper III §6 / Finding F92) obeys a closure law: a two-constituent bound state is stable at the unique angle where its unitarity phase budget closes, and the object enforcing the budget is the binder. The baryon sector obeys the *same grammar* with a different budget — the $\mathbb{Z}_3$ **centre phase** (Findings F97/F98):

Closure principle. A stable composite is a configuration whose phase budget closes exactly; the object enforcing the budget is itself the binding agent. Non-closure is priced either by over-wrap (kinematic instability, the pair sector) or linearly in separation (confinement, the colour sector).

The closure arithmetic is exact: qqq gives $3 \times \frac{2\pi}{3} = 2\pi \equiv 0$ (closed), $q\bar{q}$ gives 0 (closed), but a diquark qq gives $\frac{4\pi}{3} \not\equiv 0$ (open — confined, never asymptotic). The enforcer/binder identification is exact (Finding F98): the centre charge n is simultaneously the **budget label** (N-ality 0 $\Leftrightarrow$ asymptotic) and the **binder coefficient** (the multiplier of the dual-superconductor flux-tube tension $\sigma_{\text{BPS}} = 2\pi v^2 |n|$, Paper V §4), so $\sigma = 0$ exactly when the budget closes and $\sigma > 0$ otherwise. Only N-ality-0 combinations appear as asymptotic states — qqq , $q\bar{q}$, pentaquark, hexaquark, hybrids — a selection rule that matches every observed hadron and would be falsified by any free quark, free diquark, or fractionally-charged asymptotic state.

3.2 The no-go: baryon mass is field energy

Extending the F92 two-body fixed point to three constituents fails as an exact theorem (Finding F97). The F92 triple coincidence — amplitude unitarity, composite-mass peak, and phase wrap all meeting at 45° — holds *only* for $N = 2$: the caps $\sin(\pi/2N) = N^{-1/2}$ coincide at $N = 1, 2$ and split strictly for $N \geq 3$. For $N = 3$ a forbidden gap ($30^\circ, 35.264^\circ$) opens between the per-constituent wrap cap ($m_c \leq \frac{1}{2}$

exactly) and the unitarity cap, and both candidate three-body fixed points (trilinear $m = y^3$ and bilinear $m = y^2$) land inside it — allowed by unitarity, killed by over-wrap. **No three-constituent state can be a phase-kinematic bound state.**

This no-go is itself a successful prediction: if the baryon sat at a phase fixed point its mass would be bounded by the constituent rest-phase sum, but the measured proton has

$$\frac{2m_u + m_d}{m_p} = \frac{2(2.16) + 4.67}{938.272} = 0.96\% \quad (\text{PDG } \overline{\text{MS}}, 2 \text{ GeV}).$$

Baryon mass is $\sim 99\%$ **field energy** — set by the string tension σ and the colour-dielectric condensate, not by constituent rest mass. The split in the model matches the split in nature: the lepton sector (no colour, no confinement) is exactly where the phase chain *can* close (Paper X, Koide $Q = \frac{2}{3}$); the baryon sector is exactly where it *cannot*. The model also predicts there is **no** exact constituent-phase baryon-mass relation of the Koide type.

4. The dynamical pion: the $q\bar{q}$ pseudoscalar Goldstone

The residual nuclear force at long range is one-pion exchange, so the baryon sector needs a dynamical pion. It is delivered (Finding F103) by reusing the NJL gap + RPA ladder of Paper III with no new free physics: one coupling G generates the constituent mass via the gap equation $M = m_0 + 4GN_c N_f M I_1(M)$ and, summed in the $q\bar{q}$ ladder, fixes the meson poles $1 - 2G\Pi_M(q^2) = 0$. The pseudoscalar pole is the pion, the scalar pole its chiral partner σ .

- **Goldstone theorem (exact).** In the chiral limit ($m_0 \rightarrow 0$), $1 - 2G\Pi_{\text{PS}}(0) = 0$ is *identically* the gap equation (residual 2.3×10^{-14}), so the pion is the exact massless Goldstone ($m_\pi = 0$ to 1.6×10^{-7}).
- **Polarization split (exact).** $\Pi_S - \Pi_{\text{PS}} = -8N_c N_f M^2 K(q^2)$ to 1.8×10^{-16} .
- **Calibrated spectrum.** $m_c = 311.2$, $m_\pi = 140.5$, $f_\pi = 92.6$, $\langle \bar{q}q \rangle^{1/3} = -249.1$ MeV — matching measurement to 0.2-4.2%; the σ partner sits at $2m_c$ (ratio 1.0000).
- **GMOR.** $m_\pi^2 f_\pi^2 = -m_0 \langle \bar{q}q \rangle$ to 0.39%; the Goldstone scaling $m_\pi^2 \propto m_0$ is flat to 0.86% — the defining signature.
- **Real-space bound state.** The relative-coordinate $q\bar{q}$ bound state agrees with dense diagonalisation to 1.3×10^{-15} .

The pion is thus an anomalously light, certified dynamical Goldstone, with its mass and πNN coupling (via Goldberger-Treiman, $g_{\pi qq} f_\pi = M$ to 1.1%) ready to drive the nuclear force.

5. The deuteron: the first nucleus, bound by the pion tensor force

The deuteron is the proton-neutron bound state and the simplest nucleus. Its construction (Finding F104) generalises the two-body solver to the two coupled partial waves 3S_1 - 3D_1 of the $J^P = 1^+$, $I = 0$ channel, with the static one-pion-exchange potential

$$V_\pi(r) = -\frac{f^2}{4\pi} m_\pi [(\sigma_1 \cdot \sigma_2)Y(x) + S_{12}T(x)], \quad x = \frac{m_\pi r}{\hbar c},$$

with $Y(x) = e^{-x}/x$, $T(x) = (1 + 3/x + 3/x^2)e^{-x}/x$, and the tensor operator $S_{12} = 3(\sigma_1 \cdot \hat{n})(\sigma_2 \cdot \hat{n}) - \sigma_1 \cdot \sigma_2$ that mixes $L = 0$ and $L = 2$.

The headline result is that the deuteron binds *only* through the pion tensor force. At a fixed core radius the full 3S_1 - 3D_1 OPEP binds (-2.00 MeV) while central-only OPEP is unbound ($+0.66$ MeV): the tensor $L = 0 \leftrightarrow L = 2$ coupling supplies the missing attraction — the textbook reason the deuteron exists. The tensor strength is certified exactly: the spin-angular $\langle S_{12} \rangle$ matrix built from Clebsch-Gordan + spinor-spherical-harmonic quadrature equals the Rarita-Schwinger form $\begin{bmatrix} 0 & 2\sqrt{2} \\ 2\sqrt{2} & -2 \end{bmatrix}$ to 9.8×10^{-15} , the off-diagonal $2\sqrt{2}$ being exactly the mixing that binds. The result is a single shallow bound state, $J^P = 1^+$, $I = 0$, with a few-percent tensor-induced D-state ($P_D \approx 7\%$, vanishing when the tensor is switched off), and the 3S_1 tail reproducing $\kappa = \sqrt{M_N E_b}/\hbar c$ to 0.34% . Tuned to the physical $E_b = 2.224$ MeV, the model returns the physical $\kappa = 0.2316$ fm $^{-1}$. The carrier m_π is the Paper-III/§4 model output; the external numbers are the πNN coupling via Goldberger-Treiman (one number, g_A) and M_N (absolute scale, roadmap P2/P6).

6. The short-range NN repulsive core, derived

The deuteron build initially used a *tuned* hard core to regularise the $1/x^3$ tensor singularity — the one ingredient the roadmap flagged as not yet derived. Finding F113 supplies it from the model’s own first principles, with no new free physics.

A nucleon is a colour-singlet of three genuine CA fermions (ε_{abc} colour-antisymmetric $\times$ $SU(6)$ spin-flavour-symmetric). Two nucleons pushed together are **six identical fermions**, so the total six-quark wavefunction must be antisymmetric. Pauli antisymmetry forces the two clusters into the spatially-symmetric $[6]$ colour-spin configuration, which is **chromomagnetically unfavourable** — its colour-spin energy sits far above two free nucleons. The overlap therefore costs energy. The colour-magnetic interaction is built from the model’s own generators through exact $SU(N)$ Fierz swap identities (everything rational, avoiding the chiral-transform numerical hazard):

$$\lambda_i \cdot \lambda_j = 2P_{ij}^c - \frac{2}{3}, \quad \sigma_i \cdot \sigma_j = 2P_{ij}^s - 1, \quad H_{\text{CM}} = -\sum_{i < j} (\lambda_i \cdot \lambda_j)(\sigma_i \cdot \sigma_j) [g_{\text{cm}}].$$

The single coupling g_{cm} is fixed by the measured N - Δ splitting, $M_\Delta - M_N = 16 g_{\text{cm}} = 293 \text{ MeV} \Rightarrow g_{\text{cm}} = 18.31 \text{ MeV}$ — the *same* coupling the model already uses for baryon mass splittings, not a new parameter. The exact six-quark RGM norm kernel gives $n(R=0) = 20/9 > 0$, so the deuteron channel is **not** Pauli-forbidden (consistent with the deuteron existing), and the repulsion is the dynamical chromomagnetic one: **+341.8 MeV at zero separation** (exact given g_{cm}), falling monotonically to zero over the quark-overlap range. Re-binding the deuteron with this derived core (no hard wall) returns the physical $E_b = 2.224 \text{ MeV}$, $\kappa = 0.2316 \text{ fm}^{-1}$, a single bound 1^+ state, with the tensor still essential (16/16 PASS). The one remaining tuned quantity is the physical quark size $b \approx 0.41 \text{ fm}$, not an ad-hoc wall radius.

7. Verification summary

Result	Section	Residual / status
Colour-singlet gauge invariance $B \rightarrow \det(V)B$	§2.1	1.2×10^{-15}
Zero colour charge, $C_2 S\rangle = 0$	§2.1	1.4×10^{-16}
Proton Q, B, T_3 exact	§2.2	0 (rational)
Spin-flavour S_3 -symmetry; full Ψ antisymmetric	§2.2	0 (bit-for-bit)
Linear binding $V(R) = \sigma R$	§2.3	0 (exact area law)
$\mathbb{Z}_3$ closure ($qqq, q\bar{q} \rightarrow 0$; $qq \rightarrow 4\pi/3$)	§3.1	exact (integer)
Enforcer = binder ($\sigma = 0$ iff N-ality 0)	§3.1	exact / BPS Tier-1
Three-constituent no-go (over-wrap gap)	§3.2	exact (50 dp)
Quark-sum/ $m_p = 0.96\%$ (baryon mass = field energy)	§3.2	quantitative
Pion Goldstone (chiral limit)	§4	2.3×10^{-14}
GMOR / Goldstone scaling	§4	0.39% / 0.86%
Deuteron tensor strength $\langle S_{12} \rangle = \text{R-S}$	§5	9.8×10^{-15}
Deuteron binds only via tensor; $E_b = 2.224 \text{ MeV}$	§5	-2.00 vs $+0.66 \text{ MeV}$; 2×10^{-5}
NN core +341.8 MeV from quark Pauli + CM	§6	exact (rational, given g_{cm})

Result	Section	Residual / status
Derived-core re-binding	§6	16/16 PASS

Underlying findings: F71 (colour-singlet proton), F86/F88/F94 (confinement / condensate), F97 (centre-phase no-go), F98 (enforcer = binder), F103 (dynamical pion), F104 (deuteron / tensor force), F113 (NN repulsive core).

8. Discussion and scope

The baryon sector closes the matter content: confined colour (Paper V) produces a colour-neutral, correctly-charged, Pauli-consistent proton; a centre-phase closure principle governs which colour states are stable and *predicts* that baryon mass is field energy rather than constituent rest mass; and the residual nuclear force is built bottom-up — a dynamical pion Goldstone, the deuteron it binds through the tensor force, and a short-range repulsive core derived from quark substructure rather than tuned. The lepton/baryon contrast is structural: the phase chain closes for the colourless leptons (Koide) and provably cannot for the three-quark baryon, matching nature's split.

The honest scope is the matter-binding roadmap's remaining phases. P2 (a dynamical, real-time three-quark baryon with a measured mass, beyond the operator-level proton and energetic binding argument) is open; the proton mass and QCD scale Λ are Tier-B calibration. P5 (atoms / hydrogen — binding a proton and electron electromagnetically) and the cross-cutting P6 (SI absolute scale via the Paper-VII cell) complete the chain from substrate to nucleus to atom. None of these affects the verified structure delivered here: the proton, the closure principle, the pion, the deuteron, and the derived NN core all stand at machine precision or exact rationals.

References

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3. M. Gell-Mann, R. Oakes, B. Renner, *Phys. Rev.* **175**, 2195 (1968) — GMOR relation.
4. H. Yukawa, "On the interaction of elementary particles," *Proc. Phys.-Math. Soc. Japan* **17**, 48 (1935) — one-pion exchange.
5. M. Oka, K. Yazaki, *Prog. Theor. Phys.* **66**, 556 (1981); A. Faessler *et al.* — quark-Pauli / colour-magnetic origin of the NN repulsive core.
6. Particle Data Group, *Review of Particle Physics* (2024) — quark and hadron masses; deuteron E_b .
7. Project findings: F71, F86, F88, F94, F97, F98, F103, F104, F113.

Companion papers: III (NJL constituent mass / pion machinery), V (strong force / confinement / closure), IX (fermion content), X (lepton/baryon closure contrast).